Book

1) Physical Layer: Wireless Channel Modes

Before we just studied the standard path loss channel attenuation model, however new phenomena will change this value. The new effects will cause an oscillation around the main PL avg power:

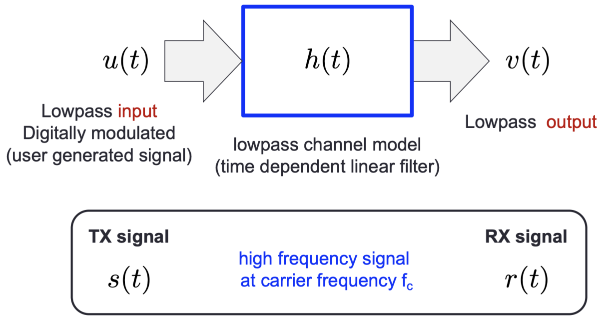
1.1) Recap

Decibel

$$P[dBW] = 10\log_{10}(\frac{P[W]}{1[W]})$$

$$P[dBm] = 10\log_{10}(\frac{P[W]}{1[mW]}) = P[dBW] + 30$$

Channel Equivalent Low Pass Model



Channel Mode

This shows that the lowpass output is represented by the TX signal as:

$$s(t) = \Re\left\{u(t)e^{j2\pi f_c t}\right\}$$

where $u(t) = x(t) + j(t)$ is a complex baseband signal and therefore the tx signal becomes

$$\begin{aligned} s(t) &= \Re\left\{(x + jy)(\cos + j\sin)\right\} = \Re\left\{x\cos - y\sin + j(x\sin + y\cos)\right\} \\ &= x(t)\cos(2\pi f_c t) - y(t)\sin(2\pi f_c t) \end{aligned}$$

The received signal is represented as

$$r(t) = \Re\left\{v(t)e^{j2\pi f_c t}\right\}$$

Supposing to know $h(t)$ to be a linear filter $h(t) = ae^{j\Delta}\delta(t-\tau)$, then:

$$v(t) = \int_{-\infty}^{t} u(t-\xi)h(\xi)d\xi$$

is the convolution with $u(t)$ that has support in $[0, \infty)$.

$$v(t) = \int_{-\infty}^{\infty} u(t-\xi)h(\xi)d\xi = ae^{j\Delta}u(t-\tau)$$

the scaling, phase and time shifts can all be reversed $\implies$ **distortion free channel!**

$$h(t) = ae^{j\Delta}\delta(t-\tau) \xrightarrow{\mathcal{F}} H(f) = ae^{j\Delta}e^{-j2\pi f\tau}$$

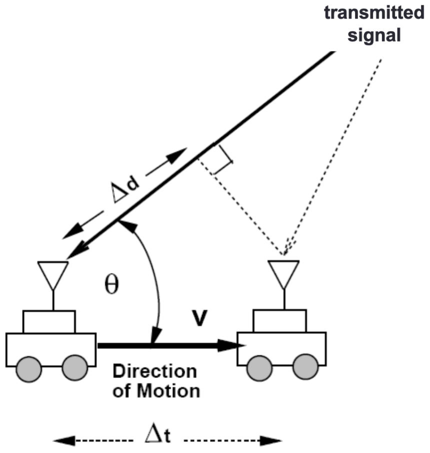
and thus

$$|H(f)| = a \quad \angle H(f) = \Delta - 2\pi f\tau$$

We want constant modulus and linear phase in the whole band of the signal (not possible in real systems).

Doppler Effect

Doppler Effect:



Doppler

Due to the movement, the path will be different

$$\Delta d = v \Delta t \cos(\theta)$$

And the doppler phase/frequency are

$$\frac{\Delta \phi}{2\pi} = \frac{\Delta d}{\lambda} \rightarrow \begin{matrix} \Delta\phi = 2\pi v \Delta t \cos(\theta)/\lambda \\ f_D = \dfrac{v\cos(\theta)}{\lambda} \end{matrix}$$

The **doppler frequency** (f_D) will be added to the carrier frequency and impacts the performance.

Path Loss

Transmit $s(t)$ with power P_t and $r(t)$ the received signal with the avg power of P_r

$$P_L = P_t/P_r \rightarrow 10\log_{10}(P_L)\ [dB]$$

In free space

$$\frac{1}{P_L} = \frac{P_r}{P_t} = \left(\frac{\sqrt{G_l}\lambda}{4\pi d}\right)^2 \rightarrow P_L[dB] = -10\log_{10}\left(\frac{G_l\lambda^2}{(4\pi d)^2}\right)$$

Thus the power falls off proportional to $1/d^2$.

In dBm the received power can be written as

$$P_r[dBm] = P_t[dBm] + 10\log_{10}(G_l) + 20\log_{10}(\lambda) - 20\log_{10}(4\pi) - 20\log_{10}(d)$$

Example.

Example 2.1: Consider an indoor wireless LAN with $f_c = 900$ MHz, cells of radius 100 m, and nondirectional antennas. Under the free-space path loss model, what transmit power is required at the access point such that all terminals within the cell receive a minimum power of 10 μW. How does this change if the system frequency is 5 GHz?

Text

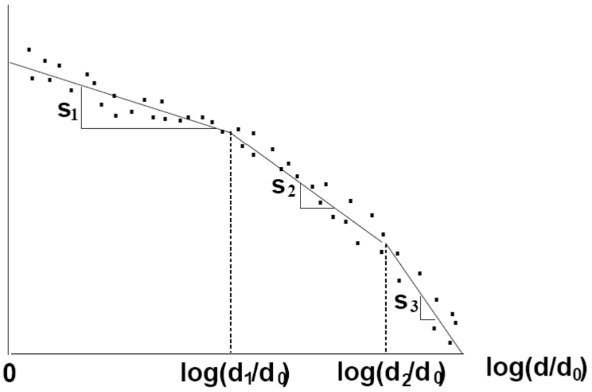
First recall that directional antennas have $G_l = 1$
By inverting the formula we obtain

$$P_t = P_r\left[\frac{4\pi d}{\lambda}\right]^2 = P_r\left[\frac{4\pi d f}{c}\right]^2 = \begin{cases} 1.45\ W \rightarrow 1.61\ dBW & f = 900\ Hz \\ 43.9\ kW \rightarrow 16.42\ dBW & f = 5\ GHz \end{cases}$$

1.2) Preliminaries

Empirical Path Loss Models

- **Okamure Model**
  This model adjusts PL with empirical data
- **Piecewise Linear (Multi Slope) Model**
  This model gathers empirical data and then uses a number N of different slopes to better approximate the curve, form here we have $N-1$ breakpoints at distances $d_j,\ j \in \{1, \dots, N-1\}$



Example (y axis is Pr dB)

In this case

$$P_r[\text{dB}] = \begin{cases} P_t + K - 10\gamma_1 \log_{10}(d/d_0) & d_0 \le d \le d_c \\ P_t + K - 10\gamma_1 \log_{10}(d_c/d_0) - 10\gamma_2 \log_{10}(d/d_c) & d > d_c \end{cases}$$

Where K and d_c are obtained via regression to fit empirical data.

A single model is difficult to obtain, therefore we use a **simplified path loss model**:

- Set K as the free space gain:

$$\boxed{K[\text{dB}] = 20\log_{10}\left(\frac{\lambda}{4\pi d_0}\right)}$$

- $$\boxed{P_r = P_t K (d_0/d)^\lambda \to P_r[\text{dBm}] = P_t[\text{dBm}] + K[\text{dB}] - 10\gamma \log_{10}(d/d_0)}$$

- And d_0 is usually given as an empirical approximation

Example.

Example 2.3: Consider the set of empirical measurements of P_r/P_t given in the table below for an indoor system at 900 MHz. Find the path loss exponent γ that minimizes the MSE between the simplified model (2.40) and the empirical dB power measurements, assuming that $d_0 = 1$ m and K is determined from the free space path gain formula at this d_0. Find the received power at 150 m for the simplified path loss model with this path loss exponent and a transmit power of 1 mW (0 dBm).

Text

Distance from transmitter d_i [m]	$M = P_r/P_t$ [dB]
10	-70
20	-75
50	-90
100	-110
300	-125

Table

Recall from the model that

$$K = 20\log_{10}\frac{c}{4\pi f d_0} = -31.54[\text{dB}]$$

and the model has the fomula:

$$M_{model}(d_i) = K - 10\log_{10} d_i[\text{dB}]$$

The MSE is obtained by

$$MSE = \frac{1}{5}\sum_{i=1}^{5}\left(M_{meas}(d_i) - M_{model}(d_i)\right)^2 = \frac{1}{5}(1571.47\gamma^2 - 11659.9\gamma + a)$$

The minimization is solved by analyzing the function:

$$MSE' = \frac{1}{5}(3142.94\gamma - 11654.9) = 0 \rightarrow \gamma^* = 3.71$$

This is a minimum since $MSE'' = \frac{1}{5}3142.94 > 0$. *Moreover the term* $\frac{1}{5}$ *can be neglected.*
Finally, put the data in the formula

$$P_r = 0 - 31.54 - 10 \cdot 3.71\log_{10}(150) = -112.27[\text{dBm}]$$

Shadowing

The shadowing can be modeled as a Gaussian Process with zero mean since the free space path loss accounts for the mean and the shadowing varies around that.

$$g(t)[dB] = \sigma_{SH} \cdot g_{SH}(t)[dB] + g_{PL}[dB]$$

The shadowing is a log-normal distribution, that is

$$\psi = \frac{P_t}{P_r} \overset{\triangle}{=} \alpha(t)^2$$

where $\alpha(t)^2$ is **the attenuation on the power** and it's sqrt the signal amplitude
It can be shown that it is gaussian distributed:

$$p(\psi_{[\text{dB}]}) = \frac{1}{\sqrt{2\pi\sigma^2_{\psi_{[\text{dB}]}}}}\exp\left\{-\frac{(\psi_{[\text{dB}]} - \mu_{\psi_{[\text{dB}]}})^2}{2\sigma^2_{\psi_{[\text{dB}]}}}\right\}$$

where $\mu_{\psi_{[\text{dB}]}}$ and $\sigma_{\psi_{[\text{dB}]}}$ are the mean and std of $\psi_{[\text{dB}]} = 10\log_{10}\psi$ (in dB) respectively

Proof:
define as log-normal distributed

$$p(\psi) = \frac{10/\ln 10}{\psi\sqrt{2\pi\sigma^2_{\psi_{dB}}}}\exp\left\{-\frac{(10\log_{10}\psi - \mu_{\psi_{dB}})^2}{2\sigma^2_{\psi_{dB}}}\right\}$$

Now transform it in dB by

$$\psi_{[\text{dB}]} = 10\log_{10}\psi \rightarrow |J| = \frac{\partial\psi}{\partial\psi_{[\text{dB}]}} = \frac{\psi\ln 10}{10}$$

By replacing and multiplying by the Jacobian we obtain the PDF in log domain

□

Exercise.

From the previous exercise now also find $\sigma^2_{\psi_{[\text{dB}]}}$

This is straight forward as it is sufficient to find the MSE:

$$MSE = \frac{1}{5}\sum_{i=1}^{5}(M_{meas}(d_i) - M_{model}(d_i))^2 = 13.29[\text{dB}]$$

In the **combined path loss+shadowing** we suppose that the shadowing produces variance around the path loss. Therefore we can set it as **a gaussian distribution with 0 mean and variance $\sigma^2_{\psi_{[\text{dB}]}}$**
And thus

$$P_r[\text{dB}] = P_t[\text{dB}] + 10\log_{10}K - 10\log_{10}(d/d_0) + \psi_{[\text{dB}]}$$

usually we have: $\mu = 0,\ K = -31.54[\text{dB}],\ \gamma = 3.71,\ \sigma = 13.29[\text{dB}]$

From here we can compute the outage:

Definition 1 (Outage)**.**

The outage is the probability for which the received power is lower than a threshold $P_{\min}$

$$P_{outage} = p_{out}(P_{\min}, d) = \text{Prob}(P_r(d) < P_{\min})$$

Which becomes

$$P_{outage} = 1 - Q\left(\frac{P_{\min} - (P_t[\text{dB}] + K[\text{dB}] - 10\gamma\log_{10}(d/d_0))}{\sigma_{\psi_{[\text{dB}]}}}\right)$$

Proof
From the definition we can rewrite it in the following way:

$$\begin{aligned} P_{outage} &= 1 - P(P_r \geq P_{\min}) \\ &= 1 - P(P_t[\text{dB}]) + K[\text{dB}] - 10\gamma\log_{10}(d/d_0) + \psi_{[\text{dB}]} \geq P_{\min} \\ &= 1 - P(\psi_{[\text{dB}]} \geq P_{\min} - (a)) \end{aligned}$$

Since $\psi_{[\text{dB}]}$ is gaussian distributed we can use the tail function $P(X \geq a) = Q(a/\sigma)$ and the proof is complete.

□

1.3) Multi-Path Fading

In real scenarios Line of Sight (LoS) is not present and the signals arrive by reflections and scattering, that is, from many directions, with different delays (τ_n) and with longer lengths (l_n)

The final signal at the receiver can be summarized as a sum of multiple signals:

$$\begin{aligned} r(t) &= \Re\{r'(t)e^{j2\pi f_c t}\} \\ r'(t) &= \sum_{N=1}^{N(t)} \alpha_n(t)e^{-j\phi_n(t)}u(t - \tau_n(t)) \\ \phi_n &\stackrel{\triangle}{=} 2\pi f_c\tau_n - 2\pi\int_0^t \frac{v(x)}{\lambda}\cos(\theta_n(x))dx \stackrel{\triangle}{=} 2\pi f_c\tau_n - \phi_{D_n}(t) \end{aligned}$$

Where ϕ_D is the doppler frequency. With a constant speed and angle (approximation for $t \to 0^+$) we have

$$\phi_n(t) = 2\pi f_c \tau_n - 2\pi \frac{v}{\lambda} \cos(\theta_n) t$$

Example.

Let $f = 1\ GHz$, then at a traveled distance of $d = 100\ m$ we have:

$$\begin{aligned} \tau_n &= \frac{d}{c} = 0.33\ \mu s \\ \theta_n &= 2\pi f \tau_n = 120\,\mathrm{deg} \end{aligned}$$

Let's focus on a longer path of $d = 110\ m$

$$\begin{aligned} \tau_n &= \frac{d}{c} = 0.36\ \mu s & \approx \tau_{n,d=100} \\ \theta_n &= 2\pi f \tau_n = 240\,\mathrm{deg} & = 2\theta_{n,d=100} \end{aligned}$$

Now let $f = 20\ GHz$ and $d = 100m$

$$\begin{aligned} \tau_n &= \frac{d}{c} = 0.33333\ \mu s \\ \theta_n &= 2\pi f \tau_n = 240\,\mathrm{deg} \end{aligned}$$

At $d = 101\ m$ the delay is roughly the same but the phase $120\,\mathrm{deg}$.

From the above example it is clear that a slight variation in distance corresponds to a slight difference in delay, but a noticable difference in phase, therefore **the phase component cannot be simplified**

Claim 2.

The low pass channel can be modeled with an LTI filter with impulse response

$$g(t, \tau) = \sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} \delta(\tau - \tau_n(t))$$

Proof

The proof is very straightforward, just convolve the impulse response and the signal, manipulate algebraically and then notice that the form is equivalent

$$\begin{aligned} r(t) = \mathrm{Re}\{(g(t,\tau) * u(t)) e^{j2\pi f_c t}\} &= \mathrm{Re}\left\{ \int_{-\infty}^{+\infty} g(t,\tau) u(t-\tau) \mathrm{d}\tau\, e^{j2\pi f_c t} \right\} \\ &= \mathrm{Re}\left\{ \int_{-\infty}^{+\infty} \sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} \delta(\tau - \tau_n(t)) u(t-\tau) \mathrm{d}\tau\, e^{j2\pi f_c t} \right\} \\ &= \mathrm{Re}\left\{ \sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} \int_{-\infty}^{+\infty} \delta(\tau - \tau_n(t)) u(t-\tau) \mathrm{d}\tau\, e^{j2\pi f_c t} \right\} \\ &= \mathrm{Re}\left\{ \sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)} u(t - \tau_n(t))\, e^{j2\pi f_c t} \right\} \end{aligned}$$

Proof

□

Definition 3 (Delay Spread)**.**

The maximum delay between the multi path signal is called delay spread:

$$\max_{i,j} |\tau_i(t) - \tau_j(t)| = \tau_m \overset{\triangle}{=} \text{Delay Spread}$$

Based on the delay spread we can distinguish two cases:

Narrowband (FLAT) Fading

Assumption.

$$\tau_m \ll B_u^{-1} \approx T_s \implies \tau_i(t) \approx \tau_j(t) \approx \hat{\tau}\ \forall i, j$$

Now the impulse response can be written as

$$g(t,\tau) = \sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\varphi_n(t)} \delta(\tau - \hat{\tau}) \overset{\triangle}{=} g(t)\delta(\tau - \hat{\tau})$$

and the impulse response becomes of linear phase:

$$g(t,\tau) \xrightarrow{\mathcal{F}} H(t,f) \overset{\triangle}{=} g(t)e^{-j2\pi f\hat{\tau}} = |g(t)|e^{j\angle g(t)}e^{-j2\pi f\hat{\tau}}$$

$$|H(t,f)| = |g(t)|$$
$$\angle H(t,f) = \angle[g(t)] - 2\pi f\hat{\tau}$$

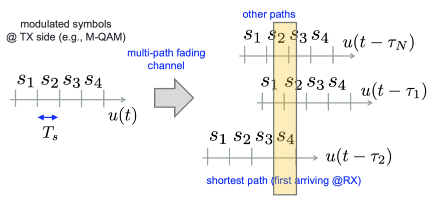
Therefore **gain is constant across frequency and the estimation of g(t) suffices to compensate for phase and amplitude displacements**

Frequency Selective (Wideband) Fading

Assumption.

$$\tau_m \gg B_u^{-1} \approx T_s$$

The signals arrive with noticeable different delays and the strongest signals (usually shortest path) arrives with ISI of other (older) signals



Example

1.4) Basic Statistical Model for the Fading Channels

2D Isotropic Scattering (Narrowband)

$$r(t) = \Re\left\{\left[\sum_{n=1}^{N(t)} \alpha_n(t) e^{-j\phi_n(t)}\right] u[t-\hat{\tau}] e^{j2\pi f_c t}\right\}$$

Here we can recognize $g(t)$. By setting $N(t) = N$ (fixed) we can write the filter in it's real and imaginary components:

$$\begin{aligned} g(t) = g_I(t) + jg_Q(t) &= \sum_{n=1}^{N} \alpha_n(t) \left[\cos(-\phi_n(t)) + j\sin(-\phi_n(t)\right] \\ &= \sum_{n=1}^{N} \alpha_n(t) \left[\cos(\phi_n(t)) - j\sin(\phi_n(t)\right] \end{aligned}$$

Rayleigh Fading

By analyzing the expectation of the real part we have:

$$E[g_I(t)] = E\left[\sum_{n=1}^{N} \alpha_n(t)\cos(\phi_n(t))\right] = \sum_{n=1}^{N} E[\alpha_n(t)]E[\cos(\phi_n(t))] = 0$$

When the phases are uniformly distributed in $[-\pi, \pi]$ (isotropic scattering) the **expectation of the cosine is 0**

Since many signals are received roughly at the same time, N is sufficiently large to use the **central limit theorem** and $X = g_I(t), Y = g_Q(t)$ are both gaussians with zero mean.

Being $G = X + jY$ it can be shown that

$$Z = |G| = \sqrt{X^2 + Y^2} \qquad p_Z(z) = \frac{z}{\sigma^2}\exp\left\{-\frac{z^2}{2\sigma^2}\right\} \qquad z \geq 0$$

Under these conditions we have

$$\boxed{\alpha(t) = Z = |g(t)|}$$

Now let's start the first order statistical description:

We want to find the **fading power** which is $|g(t)|^2 = z^2 \stackrel{\triangle}{=} \xi$

$$p_{\alpha^2}(\xi) = \frac{1}{2\sigma^2}\exp\left\{-\frac{\xi}{2\sigma^2}\right\}$$

Now let's **compare Link Budget vs SNR:**

$$P_{RX}(t) = \underbrace{\underbrace{P_{TX}}_{\text{avg trans power}} \cdot \underbrace{g_{PL}}_{\text{gain of path loss}}}_{\text{Average Received Power}} \cdot \underbrace{\alpha(t)^2}_{\text{gain of fading channel}} = P_{RX}^0 \cdot \alpha(t)^2$$

The SNR is:

$$\gamma(t) = \frac{P_{RX}^0}{N_0 B}\alpha(t)^2 = \gamma_0 \alpha(t)^2 \longrightarrow \alpha(t)^2 = \frac{\gamma(t)}{\gamma_0} \stackrel{\triangle}{=} \xi$$

and by recalling the PDF of the fading power, the PDF of the SNR becomes:

$$\boxed{p_\Gamma(\gamma) = \frac{1}{\gamma_0}\exp\left\{-\frac{\gamma}{\gamma_0}\right\}}$$

And the autocorrelation is:

$$E[g(t)g(t+\tau)] = PJ_0(2\pi f_m \tau)$$

where

$$P = \sum_{n=1}^{N} E[\alpha_n^2]$$

$$J_0(x) = \frac{1}{\pi}\int_0^{\pi} \cos(x\cos\theta)d\theta$$

J_0 is called the Bessel function of first type and order zero

The autocorrelation has zero at roughly

$$f_m\tau = \frac{v\tau}{\lambda} = \frac{l}{\lambda} \approx \frac{1}{2}$$

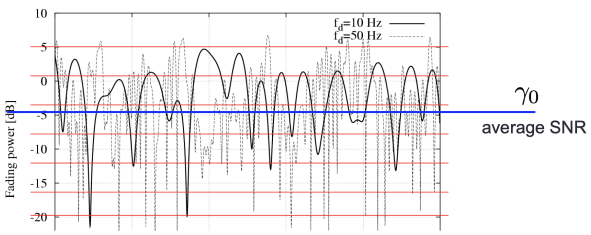
1.5) LL Error Models

First let's recap the **SNR**:

$$\gamma_0 = \underbrace{\frac{P_{tx}G_{tx}}{A_{tx}}}_{\substack{\text{transmitter}\\\text{figure}}} \cdot \underbrace{\frac{G_{rx}}{A_{rx}}}_{\substack{\text{receiver}\\\text{figure}}} \cdot \frac{1}{kB} \cdot \underbrace{\frac{1}{P_L(d,\lambda)}}_{\substack{\text{channel}\\\text{gain}}}$$

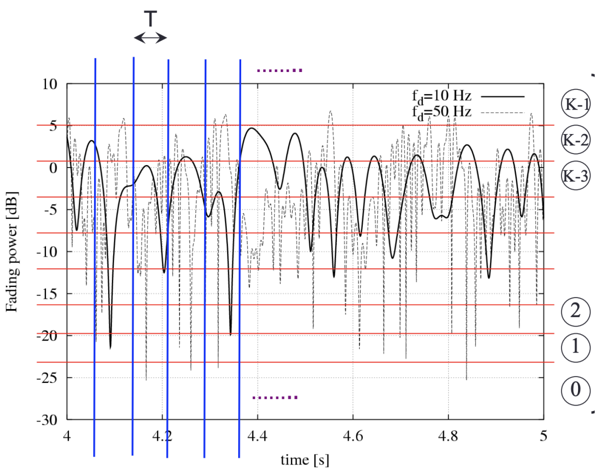
where the first two terms are the **equivalent irradiated power (EIRP)** and k the **boltzmann constant**. This is only the average SNR

However, under fading, the average SNR isn't sufficient to do a performance analysis



Example

Therefore we divide the time axis in slots of time T equivalent to a PDU transmission time (constant) and we divide it in K states with K+1 thresholds.



Example

Then, **markov chains** are used to analyze the probability to change state. For a Raleigh fading we have exponential PDF, the **the steady state probability to be in state k is**

$$\pi_k = \int_{\Gamma_k}^{\Gamma_k+1} f_R(\gamma) d\gamma = e^{-\frac{\Gamma_k}{\gamma_0}} - e^{-\frac{\Gamma_k+1}{\gamma_0}}$$

It can be shown that for a 2D isotropic rayleigh fading the **level crossing rate**, that is, the number of times the fading process crosses a set SNR level is

$$N(\Gamma_k) = \sqrt{\frac{2\pi\Gamma_k}{\gamma_0}} f_m \text{exp}\left\{\frac{-\Gamma_2}{\gamma_0}\right\}$$

Now **assume** the following:

1. the **fading is slow with respect to T**, that is $f_m T \ll 1$ therefore the SNR is approx constant in each slot $\implies$ errors iid in each block
2. **equiprobable steady state probabilities** $\pi_i = \pi_j = 1/K$ (Moayeri Model)

From 2 the thresholds can be set using this formula:

$$\Gamma_1 = \gamma_0 \text{log}\left(\frac{1}{1-\pi_0}\right) = \gamma_0 \text{log}\left(\frac{K}{K-1}\right)$$
$$\Gamma_k = \gamma_0 \text{log}\left(\frac{1}{e^{-\Gamma_{k-1}} - \pi_{k-1}}\right) = \gamma_0 \text{log}\left(\frac{K}{Ke^{-\Gamma_{k-1}/\gamma_0} - 1}\right)$$

With 1 the simples method to solve the SNR problem is with the following:

$$P_{i,j} = 0 \text{ if } |i-j| > 1$$
$$P_{k,k+1} \approx \frac{N(\Gamma_{k+1})T}{\pi_k} = KTN(\Gamma_{k+1})$$
$$P_{k,k-1} \approx \frac{N(\Gamma_k)}{T}\pi_k = KTN(\Gamma_k)$$
$$P_{k,k} = 1 - P_{k,k+1} - P_{k,k-1} = 1 - KT(N(\Gamma_{k+1}) + N(\Gamma_k))$$

Theorem 4 (LL PDU Error Probability in state k)**.**

$$P_e(k) = \int_{\Gamma_k}^{\Gamma_{k+1}} \frac{f_R(\gamma)}{P[\Gamma_k \le \gamma < \Gamma_{k+1}]} P_{PDU}(\gamma) d\gamma$$

Where $P_{PDU} = 1 - (1 - P_{bit}(\gamma))^{l_{PDU}}$ is the probability to have at least one wrong bit in the PDU

Quick recap on probability theory: An integral acts as a weighted average when used with a probability density function (this is the expected value!!!). This next example shows the probability theory approach on how to derive the formula:

Example (2-State Markov Model)**.**

In this scenario we only have two states on the markov chain and the regions are called Good (G) and Bad (B) divided by the threshold γ_{th}

Now the bit error probability in state G is directly found by recalling Theorem 4 (LL PDU Error Probability in state k).

$$\begin{aligned} P_{eG} &= P\{\text{PDU is erroneous} \mid \text{Mc in state G}\} \\ &= P\{\text{PDU wrong} \mid \gamma \ge \gamma_{th}\} \\ &= \int_{\gamma_{th}}^{\infty} \frac{f_R(x)}{P\{\gamma \ge \gamma_{th}\}} \left[1 - (1 - P_{bit}(x))^{l_{PDU}}\right] dx \end{aligned}$$

Same for P_{eB} *just with* $\int_0^{\gamma_{th}}$
The probability is the CDF of a complex rv, that is

$$P_R[a < \gamma < b] = e^{-\frac{a}{\gamma_0}} - e^{-\frac{b}{\gamma_0}}$$

We can set the γ_{th} *in order to have equiprobable steady state probabilities* $\pi_G = \pi_B = \frac{1}{2}$

$$\frac{1}{2} = P_R[\gamma \geq \gamma_{th}] = e^{-\frac{\gamma_{th}}{\gamma_0}} \rightarrow \gamma_0 \log(2)$$

2) ARQ vs HARQ

HARQ is a ARQ that also implements coding. In fact, each segment of the packet will be encoded. Let's start with a recap of coding theory.

The **encoder** encodes a input x of length k into an output y of n bits. The code rate is the percentile of the og message that gets delivered every T, that is: $r_c = k/n$. There are exactly 2^k valid codewords, one for each input message, but encoded messages live in a bigger space, that is 2^n. The **hamming distance** is the number of different bits between two messages.

$$\underbrace{y}_{n\times 1} = \underbrace{G}_{n\times k} \cdot \underbrace{x}_{k\times 1}$$
$$\text{Code Rate: } r_c = k/n$$
$$\underbrace{2^n}_{\substack{\text{encoded}\\\text{messages}}} - \underbrace{2^k}_{\substack{\text{valid}\\\text{messages}}} = \text{additional strings}$$
$$\text{Hamming Distance: } d(y,z) = \sum_{i=0}^{n-1} y_i \otimes z_i$$

In this course we consider **Reed Salomon (RS) codes** which has the following properties

$$\text{Minimum Codeword Distance: } d_{min} = N - K + 1$$
$$\text{Error Correction Capability: } t = \left\lfloor \frac{d_{min}-1}{2} \right\rfloor = \left\lfloor \frac{N-K}{2} \right\rfloor$$

Under IID assumptions we have that

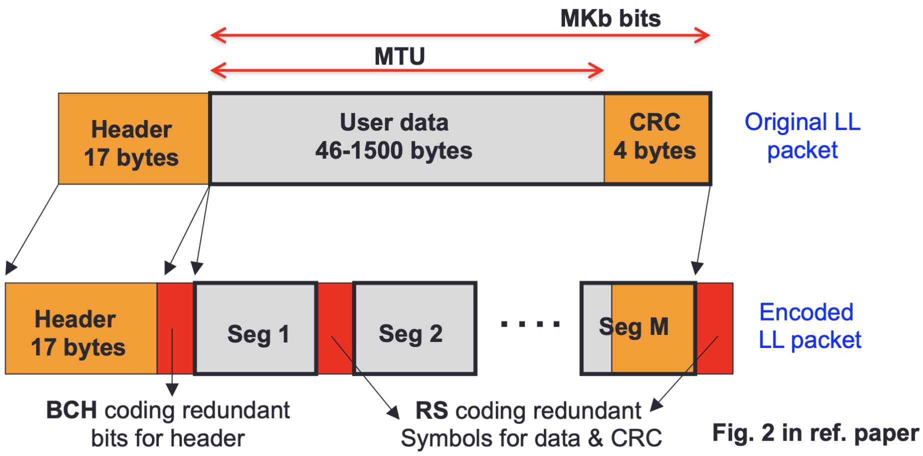
$$\text{Symbol Error Probability: } P_S = 1 - (1 - P_{bit})^b$$
$$\text{Decoder Error Probability: } P_T = \sum_{i=t+1}^{N} \binom{N}{i} P_S^i (1-P_s)^{N-i} \text{ (more than t symbols are wrong)}$$
$$\text{Total Error Probability: } P_T = P_E + P_F$$
$$P_E \text{ Upper Bound: } P_e \leq P_T \sum_{i=0}^{t} \binom{N}{i} (2^b - 1)^{i-(n-K)}$$

We also use **BCH** codes for the header

$$\text{Error Correction Capability: } t = \left\lceil \frac{d_{min}-1}{8} \right\rceil = \left\lceil \frac{N-K}{8} \right\rceil$$
$$\text{Header Error Probability: } P_h = \sum_{e=t+1}^{n} \binom{n}{e} P_{bit}^e (1-P_{bit})^{n-1}$$

We also consider the difference between two types of error cases:

- **Bursty Errors (Best Case):** all bits are wrong, therefore all of them can be corrected
- **IID Errors (Worst Case):** only t bits can be corrected



Packet Fragmentation

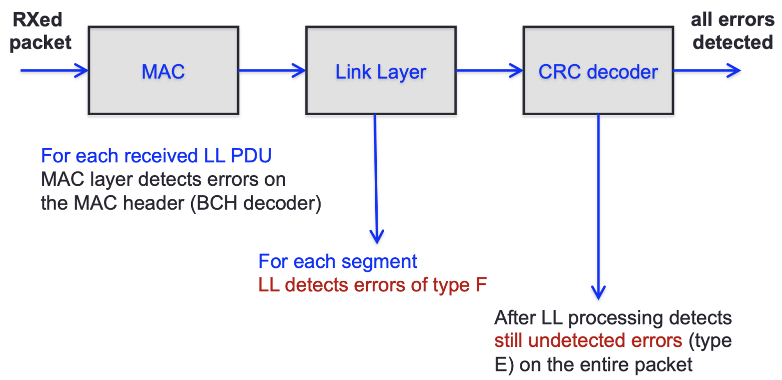
Here are the assumptions:

- **Header** uses BCH coding (only iid errors), more performant
- **Data Segments** use RS coding
- Each segment is one codeword
- The last Segment includes CRC check

We can identify 3 cases:

- **Correct Decoding:** The codeword is correctly decoded into the original message. (errors $\leq t$)
- **F error, Decoding Failure:** The codeword is corrupted and no valid codeword is found (errors > t)
- **E error, Decoding Error:** The codeword is decoded into the wrong message (errors < t, d(C,C')>t)

The **E error cannot be detected** by the RS codes, but only by the CRC. This means that it is not possible to identify the wrong segment, we only know that the packet is wrong, a full RETX is required.



LL Packet Detection Flow

Three types of HARQ are defined:

- **Type I:** One message (M=1)
- **Type II:** Many messages (M>1) and RETX packet if any segment has E or F errors
- **Type III:** Many messages (M>1) and REXT segment if F error, RETX packet if E error. It is possible that a RETX segment will contain an E error.

2.1) Throughput

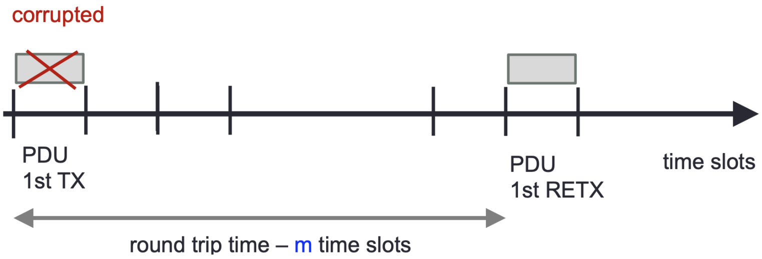
First define:

- **Average Number Of Transmitted Bits To Successful LL PDU Delivery:** $E[N^i_{tr}]$ (i is the scheme $i = 1, 2, 3$)
- **Throughput:**

$$\eta = \frac{\text{num of information bits in packet}}{E[N^i_{tr}]}$$

SR-ARQ

Now consider an **IDEAL SR-ARQ** with feedback after m time slots



SR_ARQ

The **avg number of transmissions needed** is:

$$E[N_{tx}] = 1 + pE[N_{tx}] + (1-p)\cdot 0 \rightarrow E[N_{tx}] = \frac{1}{1-p}$$

Where 1 is the transmission of first PDU, then we need 0 retx if there is no error (1-p), but we need to **statistically restart the connection** if there is an error (p).

The packet error rate is

$$P_R^{no} = 1 - \underbrace{(1-P_h)}_{\text{header OK}}(1-P_b)^{\overbrace{MTU+CRC}^{\text{PDU size}}}$$

We also define two lengths:

- L_{inf} = MTU+CRC-C (C is CRC+IP)
- L_{pkt} = MTU+CRC+H+O (O is bits added at PHY)

And the number of bits transmitted on avg. is

$$E[N^{no}_{tr}] = L_{pkt} + P_R^{no}E[N^{no}_{tr}] \rightarrow E[N^{no}_{tr}] = \frac{L_{pkt}}{1-P_R^{no}}$$

and so we obtain the throughput as

$$\eta^{no} = \frac{L_{inf}}{E[N^{no}_{tr}]} = \frac{(MTU + CRC - C)(1 - P_R^{no})}{L_{pkt}}$$

HARQ-I

The packet error rate equals to the probability to retx, Moreover $L_{pkt}(\cdot)$ is the bits needed to TX a frame (HARQ-II, III tx different number of frames)

$$P_R^I = 1 - (1-P_h)(1-P_T)$$
$$E[N^I_{tr}] = L_{pkt}(1) + P_R^I E[N^I_{tr}] \rightarrow E[N^I_{tr}] = \frac{L_{pkt}(1)}{1-P_R^I}$$
$$\eta^I = \frac{Kb - C}{E[N^I_{tr}]}$$

Remark.

The number of bits to TX a frame is:

$$L_{pkt}(M) = MNb + H + O$$

HARQ-II

Here we must also **account for the probability to get e errors and f failures among the M segments**

$$p_{e,f}(M) = \binom{M}{e}\binom{M-e}{f} P_E^e P_F^f (1 - P_E - P_F)^{M-e-f}$$

$$P_R^I = 1 - (1 - P_h)p_{0,0}(M)$$

$$E[N_{tr}^{II}] = L_{pkt}(M) + P_R^{II} E[N_{tr}^{II}] \to E[N_{tr}^I] = \frac{L_{pkt}(M)}{1 - P_R^{II}}$$

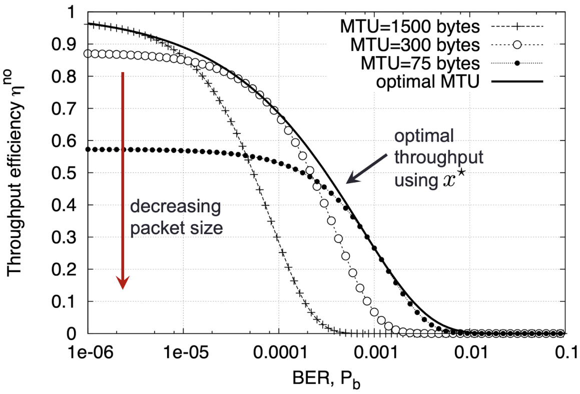
$$\eta^I = \frac{MKb - C}{E[N_{tr}^I]} = \frac{(MKb - C)(1 - P_R^{II})}{L_{pkt}(M)}$$

How do we find the **max PDU size?** Find how to maximiza the throughput (B_{LL} is the bitrate at LL)

$$\eta(x) = B_{LL}\left(\frac{x - o}{x}\right)(1 - P_b)^x$$

which results in

$$x = \frac{o + \sqrt{o^2 - \frac{4o}{\log(1-P_b)}}}{2}$$



ARQ Performance

HARQ-III

We have 5 **mutually exclusive** cases

1. Wrong Header: RETX PDU
2. NO error (0 E, 0 F): END, no RETX
3. No decoding failure (e E, 0 F): RETX PDU
4. No decoding error (0 E, f F): RETX f segments, it is possible they contain E
5. Both errors (e E,f F): REXT f segments and then, if CRC fails again RETX

$$E[N_{tr}^{III}] = \underbrace{L_{pkt}}_{\text{PDU bits}} + \left(\underbrace{P_h}_{\text{Case 1}} + \underbrace{(1-P_h)\sum_{e=1}^{M} p_{e,0}(M)}_{\text{Case 3}} \right) E[N_{tr}^{III}] +$$
$$+(1-P_h)\underbrace{\left(\sum_{f=1}^{M} p_{0,f}(M)\left(A(f)+B(f)E[N_{tr}^{I}II]\right)\right)}_{\text{Case 4}} +$$
$$+\underbrace{\sum_{f=1}^{M-1}\sum_{e=1}^{M-f} p_{e,f}(M)(A(f)+E[N_{tr}^{III}])}_{\text{Case 5}} + \underbrace{(1-P_h)p_{0,0}(M)}_{\text{Case 2}} \cdot 0$$

Let

- $A(f)$ be the number of TX bits over LL until all f segments are sent correctly
- $B(f)$ be the probability that at least one segment has E error

In fact:

- Case 1: Header error$\rightarrow$ restart connection
- Case 2: No error, so stop process ($\cdot 0$)
- Case 3: Header is wrong $(1-P_h)$ $\rightarrow$ restart connection with probability of 0 F errors
- Case 4: Header is wrong $(1-P_h)$, for each probability of having 0 E errors and f F errors $A(f)$ bits are sent that have $B(f)$ probability to restart connection
- Case 5: first notice $B(f) = 1$ since we are sure at least one E error happened. TODO

It is possible to rewrite it as:

$$E[N_{tr}^{III}] = \frac{L_{pkt}(M) + (1-P_h)\left(\sum_{f=1}^{M}\sum_{e=0}^{M-f} p_{e,f}(M)A(f)\right)}{(1-P_h)\left(1-\sum_{f=1}^{M} p_{0,f}(M)B(f) - \sum_{f=0}^{M-1}\sum_{e=1}^{M-f} p_{e,f}(M)\right)}$$

3) Exercises in Class

Warmup Exercise 1

Let a wireless link between a TX and a RX have following specifications:

- PDU Length at LL: PDU_l
- Bit Error Rate over Wireless Link: P_b
- Max # of TX per PDU: L
- Assume iid Errors
  Find 1) the packet error rate of RX P_{PDU} (between PHY and LL) and 2) the residual error rate at RX P_{LL} (between LL and NET)

1. This is straight forward as $P_{PDU} = 1 - (1 - P_b)^{PDU_l}$
2. A bit more cumbersome as we must find the probability that L TX are wrong.
   Let X be the rv that counts the # of consecutive unsuccessful TX, this is $X \sim \text{Geom}(P_{PDU})$
   Then, we must find

$$
\begin{aligned}
P[X \geq L] = 1 - P[X < L] = 1 - \sum_{n=0}^{L-1} P[X = n] \\
&= 1 - \sum_{n=0}^{L-1} P_{PDU}^n (1 - P_{PDU}) = 1 - (1 - P_{PDU}) \frac{1 - P_{PDU}^L}{1 - P_{PDU}} \\
&= P_{PDU}^L
\end{aligned}
$$

Warmup Exercise 2

Let a wireless link between a TX and a RX have following specifications:

- PDU Length at LL: PDU_l
- Bit Error Rate over Wireless Link: P_b
- Max # of TX per PDU: L
- Assume iid Errors
- $PDU_l = H + PAY$ **all known**
- **The TX time at LL:** t_F
  Find 1) B_{LL}^T, B_{LL}^I, the bitrates at the LL for the total PDU and Information bits in the cases where $L \in \mathbb{N}$ and $L \to \infty$.

1. First Notice that $B_{LL} = \mathbb{E}[\text{data}]/\mathbb{E}[T]$
   And define the following RV:

- X: rv that counts # of succesfull bits TXed
- Y: rv that counts # of successfull information bits TXed
- T: rv that counts tot time to send LL PDU

Notice that X and Y are similar, only with different amounts of size, therefore all results for X can be translated directly to Y by changing PDU_l to $PDU_l - H$.

First find $\mathbb{E}[X], \mathbb{E}[Y]$

- In the case $L \in \mathbb{N}$ we have
  Notice that X derives from a bernoulli rv $B \sim \text{Bernoulli}(P_{LL})$ and $X = PDU_l B$ where B states the probability to successfully TX a PDU in L tries.

$$
\begin{aligned}
\mathbb{E}[X] &= PDU_l \cdot \mathbb{E}[B] = PDU_l \sum_{n=0}^{1} b_i P_B(b_i) \\
&= PDU_l(0 \cdot P_{LL} + 1 \cdot (1 - P_{LL})) = PDU_l(1 - P_{PDU}^L)
\end{aligned}
$$

And for the information bits just $\mathbb{E}[Y] = (PDU_l - H)(1 - P_{PDU}^L)$

- In the case $L \to \infty$ we have
The PDU clearly gets delivered sometimes and by logic it is clear that $\mathbb{E}[X] = PDU_l$.
It is also possible to find from the limit of the previous result $\lim_{L\to\infty} \mathbb{E}[X] = PDU_l(1 - 0)$.
The more formal approach is to recalculate P_{LL} and redo the full calculations:

$$
P_{LL} = 1 - (1 - P_{PDU}) \sum_{n=0}^{\infty} P_{PDU}^n = 1 - (1 - P_{PDU}) \frac{1}{1 - P_{PDU}} = 0
$$

and thus $B \sim \text{Bernoulli}(0)$ and therefore $\mathbb{E}[X] = PDU_l, \mathbb{E}[Y] = PDU_l - H$.

Now we must find $E[T]$ which is more cumbersome:

- In the case $L \in \mathbb{N}$ we have
Now $T = t_F G_L + L t_F B$ since with probability P_{LL} it will not TX in L tries and therefore it will take time $L t_F$.
Moreover G_L is $G \sim \text{Geom}(P_{PDU})$ that counts the number of tries until first success truncated at L # of tries.
Now

$$
\begin{aligned}
\mathbb{E}[T] &= t_F \mathbb{E}[G] + L t_F \mathbb{E}[B] \\
&= t_F \sum_{0}^{L-1} (n+1) P_{PDU}^n (1 - P_{PDU}) + L t_F P_{LL} \\
&= t_F \frac{1 - (L+1) P_{PDU}^L + L P_{PDU}^{L+1}}{(1 - P_{PDU})^2} (1 - P_{PDU}) + L t_F P_{PDU}^L \\
&= t_F \left(\frac{1 - (L+1) P_{PDU}^L + L P_{PDU}^{L+1}}{(1 - P_{PDU})} + L P_{PDU}^L \right)
\end{aligned}
$$

- In the case $L \to \infty$ we have
Again T derives from $T = t_F G$ with $G \sim \text{Geom}(P_{PDU})$ where G is the # of tries until first success
Now

$$
\begin{aligned}
\mathbb{E}[T] &= t_F \mathbb{E}[G] = t_F \sum_{n=0}^{\infty} (n+1) P_{PDU}^n (1 - P_{PDU}) \\
&= t_F (1 - P_{PDU}) \left(\sum n P_{PDU}^n + \sum P_{PDU}^n \right) \\
&= t_F (1 - P_{PDU}) \left(\frac{P_{PDU}}{(1 - P_{PDU})^2} + \frac{1}{1 - P_{PDU}} \right) \\
&= t_F \left(\frac{P_{PDU}}{1 - P_{PDU}} + 1 \right) = \frac{t_F}{1 - P_{PDU}}
\end{aligned}
$$

If we take the limit of the finite case we have the result obtained above

$$
\lim_{L\to\infty} \mathbb{E}[T] = \frac{t_F}{1 - P_{PDU}}
$$

Remark (Two Geometric RV exist)**.**

Two geometric rv exist based on what we count. We set L as the amount of times the rv was run and p the failure probability.

- **Count # of unsuccessful tries** $\in \{0, \dots, N\}$

$$P[G_{failures} = k] = p^k(1-p) \qquad \mathbb{E} = \sum_{n=0}^{L-1} np^n(1-p) \overset{L\to\infty}{\longrightarrow} \frac{p}{1-p}$$

- **Count of tries needed until first success** $\in \{1, \ldots, N\}$

$$P[G_{success} = k] = p^{k-1}(1-p) \qquad \mathbb{E} = \sum_{n=1}^{L} np^{n-1}(1-p)$$

alternatively it is also presented as

$$\mathbb{E} = \sum_{n=0}^{L-1} (n+1)p^n(1-p) \overset{L\to\infty}{\longrightarrow} \frac{1}{1-p}$$

which is mathematically equivalent.
For $L \to \infty$ we can define $G_{success} = G_{failures} + 1$ and thus $\mathbb{E}[G'] = 1 + \mathbb{E}[G]$